

Aflevering 3 - Mikkel Moth Billing

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2. Consider the function $f : [0, \pi] \rightarrow \mathbb{R}$ given by $x \in [0, \pi] \mapsto f(x) := \cos(x)$. Extend f to an odd function on the interval $[-\pi, \pi]$. Write down this odd extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$ and represent f_o graphically.

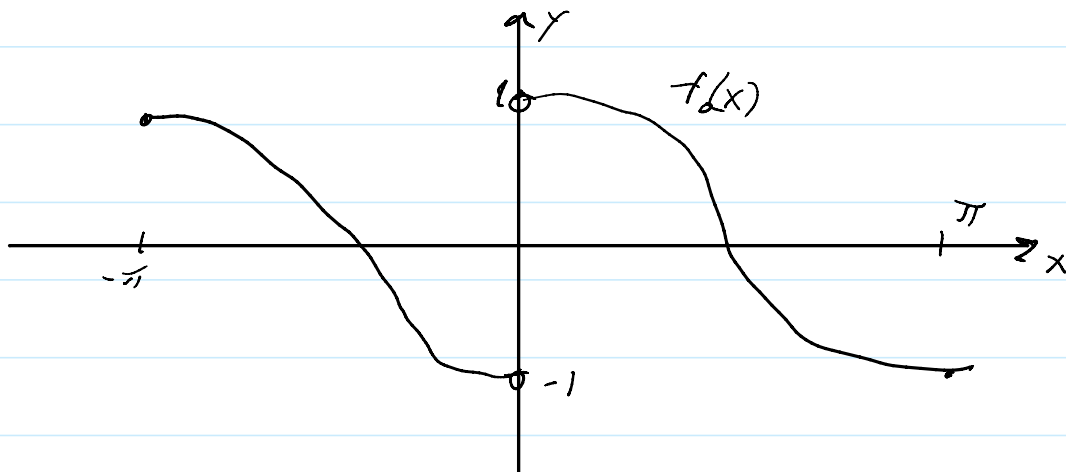
To be an odd function $f_o(x)$ has to satisfy:

$$f_o(-x) = -f_o(x)$$

$$\Rightarrow \cos(-x) = -\cos(x)$$

Since $\cos(x)$ is an even function, that is: $\cos(x) = \cos(-x)$, the odd extension must be:

$$f_o(x) = \begin{cases} \cos(x) & 0 \leq x \leq \pi \\ -\cos(x) & -\pi \leq x < 0 \end{cases}$$



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3. Compute the real Fourier sine coefficients of f defined in 2. above and write down its associated real Fourier sine series on the interval $[0, \pi]$.

We use Theorem 1.8, that states that there is no sine parts since $f(x) = \cos(x)$ is an even function.

4. Write down the points of continuity and the points of discontinuity of the extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$ found in 2. above. Is f_o continuous on $[-\pi, \pi]$? Is f_o piecewise smooth on $[-\pi, \pi]$? Explain your answers.

At the point $x=0$ f_o is discontinuous else it is continuous.
It is not continuous

It is piecewise smooth, because it is piecewise continuous and so is the derivative.

5. Compute the real Fourier series $F_o : [-\pi, \pi] \rightarrow \mathbb{R}$ of the extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$.

We use Theorem 1.8 again with $a=-\pi$, but now the function is odd, so $a_0 = a_k = 0$

$$\begin{aligned} b_k &= \frac{2}{\pi} \int_0^{\pi} \cos(x) \sin(kx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} \left(\sin[(k+1)x] + \sin[(k-1)x] \right) dx \end{aligned}$$

$$= \frac{1}{\pi} \left[\frac{-1}{k+1} \cos[(k+1)x] \right]_0^{\pi} + \left[\frac{-1}{k-1} \cos[(k-1)x] \right]_0^{\pi}$$

Since k is an integer we get:

$$= \frac{1}{\pi} \left(\frac{1}{k+1} (-1)^k + \frac{1}{k+1} \right) + \frac{1}{k-1} (-1)^k + \frac{1}{k-1}$$

$$= \left(\frac{1}{\pi(k+1)} + \frac{1}{\pi(k-1)} \right) (-1)^k + \frac{1}{\pi(k+1)} + \frac{1}{\pi(k-1)}$$

So we get the real Fourier series to:

$$F_o(x) = \sum_{k=1}^{\infty} \left[\left(\frac{1}{\pi(k+1)} + \frac{1}{\pi(k-1)} \right) (-1)^k + \frac{1}{\pi(k+1)} + \frac{1}{\pi(k-1)} \right] \sin(kx)$$

6. Using Theorems 1.22 and 1.28 (from the book [BoNa] and my lecture notes) explain why and write down the points $x \in [-\pi, \pi]$ for which the real Fourier series $F_o(x)$ converges, respectively does not converge, to $f_o(x)$. For the cases where $F_o(x)$ does not converge to $f_o(x)$, write down the exact value with respect to f_o to which $F_o(x)$ converges. (One needs to extend f_o to a 2π -periodic function on $\mathbb{R}$ in order to apply the mentioned Theorems).

We extend $f_o(x)$ to a 2π -periodic

$$f_e(x) = f_o(x + 2\pi n) = \begin{cases} \cos(x + 2\pi n) & 0 < x \leq \pi \\ -\cos(x + 2\pi n) & -\pi \leq x < 0 \end{cases}$$

The derivative of f_e is defined on $x \in [-\pi, 0)$ and $x \in (0, \pi]$ so according to 1.22 F_e converges to f_e in that interval. But for $x=0$ we use 1.28 since it is left-right differentiable. So F_e converges to:

$$\frac{f(x+0) + f(x-0)}{2} = \frac{1 + (-1)}{2} = 0$$

7. Is the extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$ an element of $L^2([-\pi, \pi])$? If yes, explain your answer, and then compute the L^2 -norm of f_o in $L^2([-\pi, \pi])$.

We find out by calculating:

$$\|f_o\|_{L^2} = \int_{-\pi}^{\pi} |\cos(x)|^2 dx$$

$$= \int_{-\pi}^{\pi} \frac{1}{2} + \frac{1}{2} \cos(2x) dx$$

$$= \left[\frac{x}{2} + \frac{1}{4} \sin(2x) \right]_{-\pi}^{\pi}$$

$$= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi < \infty$$

So it is defined in the L^2

8. Write down the definition of uniform convergence with respect to the interval $[-\pi, \pi]$.

Theorem 1.30 The Fourier series of a continuous, piecewise smooth 2π -periodic function $f(x)$ converges uniformly to $f(x)$ on $[-\pi, \pi]$.

Theorem 1.30

9. Does the real Fourier series $F_o : [-\pi, \pi] \rightarrow \mathbb{R}$ of the extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$ converge uniformly to the function f_o on the interval $[-\pi, \pi]$? Explain carefully your answer. Does the Gibbs phenomenon appear? If yes, in which points?

No, it is not continuous and therefore doesn't satisfy Theorem 1.30.

Since f_o is discontinuous in $x=0$, we see the Gibbs phenomenon here.

10. Write down the definition of convergence in L^2 -norm with respect to the space $L^2([-\pi, \pi])$.

Theorem 1.35:

Theorem 1.35 Suppose f is an element of $L^2([-\pi, \pi])$. Let

$$f_N(x) = a_0 + \sum_{k=1}^N a_k \cos(kx) + b_k \sin(kx)$$

where a_k and b_k are the Fourier coefficients of f . Then f_N converges to f in $L^2([-\pi, \pi])$, that is, $\|f_N - f\|_{L^2} \rightarrow 0$ as $N \rightarrow \infty$.

Theorem 1.35 also holds for the complex form of Fourier series.

11. Does the real Fourier series $F_o : [-\pi, \pi] \rightarrow \mathbb{R}$ of the extension $f_o : [-\pi, \pi] \rightarrow \mathbb{R}$ converge in L^2 -norm to the function f_o with respect to $L^2([-\pi, \pi])$? Explain carefully your answer.

Since we showed that f_o is an element in $L^2([-\pi, \pi])$, and F_o is the Fourier series, we know it converges in the L^2 -norm from Theorem 1.35.

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